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B.Tech. DEGREE EXAMINATION, NOVEMBER 2015.

Fifth Semester

Computer Science And Engineering

OPERATING SYSTEMS

(2013 –2014 Onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions,

All questions carry equal marks.

1. Define distributed system.
2. Name any three components of the system.
3. Define thread.
4. What is monitor?
5. Define paging.
6. Define deadlock.
7. What is the principle of optimality?
8. Define thrashing.

9. What is kernel?
10. Define physical record and logical.

PART B — ($5 \times 11 = 55$ marks)

Answer ONE questions from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the service provided by the operating system.

Or

12. (a) Explain the different types of Operations on process. (7)
- (b) Write short notes on Inter-process communication. (4)

UNIT II

13. (a) Explain briefly about round robin Scheduling with diagram. (6)
- (b) What are the criteria for scheduling? (5)

Or

14. Explain briefly about semaphores.

UNIT III

15. Describe the Banker's algorithm for dead lock avoidance.

Or

16. What is paging? Explain briefly.

UNIT IV

17. What is page replacement? Explain how it will be done in the memory?

Or

18. (a) What is thrashing? Explain.
- (b) Explain the different file access methods.

UNIT V

19. Describe briefly about disk scheduling.

Or

20. Discuss on file system structure in the linux system.

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B.Tech. DEGREE EXAMINATION,
NOVEMBER 2017.

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Computer Science and Engineering

OPERATING SYSTEMS

(2013-14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. Distinguish between multiprocessor system and distributed system?
2. What are system calls?
3. What are the different CPU scheduling algorithms?
4. Define semaphores.
5. Why do deadlocks occur in operating systems?
6. What is segmentation?
7. Why does thrashing occur in operating systems?

8. How is a directory structure used in file systems?

9. What is Disk management?

10. What is the use of a swap space?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE question from each Unit.

11. (a) Explain the scheduling algorithms in operating systems.

Or

(b) Discuss how inter-process communication takes place in operating systems.

12. (a) Explain multiple process scheduling and real-time scheduling in operating systems.

Or

(b) What is a critical region? Explain how a critical region can be implemented.

13. (a) Explain how deadlocks are handled in operating systems.

Or

(b) Discuss the concept of segmentation with paging in operating systems.

14. (a) Discuss page replacement algorithms.

Or

(b) Explain File Sharing and protection in operating systems.

15. (a) Discuss free space management and allocation methods in file systems.

Or

(b) Discuss the disk scheduling algorithms for storage management.

20. Suppose that a disk drive has 5,000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order, is :
(11)
86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130

Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk-scheduling algorithms?

- (a) FCFS
- (b) SSTF
- (c) SCAN
- (d) LOOK.

UNIT II

13. Consider the following set of processes, with the length of the CPU burst given in milliseconds. (11)

Process	Burst Time	Priority
P1	10	3
P2	1	1
P3	2	3
P4	1	4
P5	5	2

The processes arrived in the order P1, P2, P3, P4 and P5 at time 0.

- (a) Draw Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-preemptive priority (a smaller priority number implies higher priority)
- (b) What is the turnaround time of all processes for each scheduling algorithms?

Or

14. Explain the structure of monitors. Give the solution for Dining-Philosophy problem using monitor. (11)

UNIT III

15. Consider the following snapshot of the system. (11)

	Allocation			Request			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	0	0	0	0	0	0
P1	2	0	0	2	0	2			
P2	3	0	3	0	0	0			
P3	2	1	1	1	0	0			
P4	0	0	2	0	0	2			

- (a) Is the system safe?
 (b) What will happen that the process p2 makes one additional request for an instance of type C?

Or

16. (a) Consider the following segment table : (4)

Segment	Base	Length
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	96

What are the physical addresses for the following logical addresses?

- (i) 0,430
 (ii) 1,10
 (iii) 2,500

- (b) Compare the main memory organization schemes of contiguous memory allocation, pure segmentation, and pure paging with respect to the following issues :

- (i) external fragmentation
 (ii) internal fragmentation
 (iii) ability to share code across processes

UNIT IV

17. What is page replacement? Write down the basic page replacement algorithm. How many page faults occur for FIFO algorithm for the following reference string, with four page frames (Reference String : 1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 8, 1, 2) (11)

Or

18. What is Thrashing? What are the causes of Thrashing? Explain about various impacts on CPU performance by Thrashing. (11)

UNIT V

19. Explain about various methods used for allocating disk space for files. Illustrate the advantages and limitations of each method. (11)

Or

UNIT V

19. (a) Explain the structure of the file system in Linux and windows systems. (6)
(b) Write short notes on caching. (5)

Or

20. Explain the process management in Linux system. (11)

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B.Tech DEGREE EXAMINATION,
MAY 2018.

Fifth Semester

Computer Science and Engineering

OPERATING SYSTEMS

(2013-14 onwards)

Time : Three hours
Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What are the objectives of an operating system?
2. What are the five categories of system calls?
3. Define thread.
4. What is semaphore?
5. What are the three methods that deal with the deadlock problem?

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6. Write short notes on swapping.

7. Define virtual memory.

8. What is thrashing?

9. What is internal fragmentation?

10. Define buffer in I/O system?

PART B — (5 × 11 = 55 marks)

Answer ALL question ONE from each Unit.

All questions carry equal marks.

UNIT I

11. (a) Explain the differences between operating system for mainframe computers and personal computers? (6)

(b) Why are distributed systems desirable? (5)

Or

12. (a) Explain the possible operations that could be applied on process in detail. (5)

(b) What is IPC? Explain (6)

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UNIT II

13. (a) Explain briefly about round robin scheduling with example. (1)

(b) What is binary semaphore? How will you implement it? (1)

Or

14. What is critical section problem? How will you find the solution for it? (1)

UNIT III

15. Discuss briefly about Bankers algorithm with example. (1)

Or

16. What is paging? Explain briefly. (1)

UNIT IV

17. What is meant demand paging? Explain how the allocation of frames can be done? (1)

Or

18. (a) Explain the different file access methods of file management. (1)

(b) Explain the operations performed on directory. (1)

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UNIT III

15. At any instant of time, say current state has 3 process A, B, C. total 10 resources are there. Figure shows current allocation state.

Process	Has	Max
A	3	9
B	2	4
C	2	7

Free resources 3 are there. Is there exist a safe sequence of allocations that allows process to complete?

Or

16. Explain segmentation with example.

UNIT IV

17. Explain FIFO and LRU page replacement algorithm with example.

Or

18. What are the various file allocation methods? Explain with example.

UNIT V

19. Write notes on Linux kernel architecture. Advantages of Linux over windows OS.

Or

20. Given the following queue -- 95, 180, 34, 119, 11, 123, 62, 64 with the Read-write head initially at the track 50 and there are 200 cylinders numbered from 0 - 199. Discuss the following Disk scheduling algorithms (a) FCFS (b) SSTF (c) C-SCAN.

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2018.

Fifth Semester

Computer Science and Engineering

OPERATING SYSTEMS

(2013 - 14 onwards batches)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

All questions carry equal marks.

1. What is a distributed system?

2. Mention any one method for Inter process communication.

3. Define Critical section.

4. Write about real time scheduling.

5. Write the three ways to deal the deadlock problem.

6. What is paging?
7. What is demand paging?
8. What is mounting?
9. Draw the Linux OS directory structure.
10. How do you manage swap space?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Write notes on (a) Multi-processor systems
(b) Real time systems (c) System Calls.

Or

12. Discuss about process states and Process control block.

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UNIT II

13. Consider following process with length of CPU burst time in milliseconds:

Process	Burst Time
P1	6
P2	10
P3	3
P4	4
P5	2

- (a) Draw Gantt Charts illustrating execution of these processes for round robin scheduling (quantum=2)& FCFS.
 - (b) Calculate waiting time for each process for each scheduling algorithm.
 - (c) Calculate average waiting time for each scheduling algorithm.
- Consider all processes arrived in order P1, P2, P3, P4, P5 at time zero.

Or

14. What is semaphore? How do you solve dining philosopher problem using semaphore?

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